

CENTRE OF GRAVITY –
BALANCING ACTS



$$1 + 2 + 4$$
$$5 + 3 + 5$$



Riya's Big Question



Riya sees a tightrope walker at the mela. She watches with wide eyes as he walks on a thin rope high above the ground!

"How does he stay up there without falling?"

What is the secret of balancing?





The Magic Balancing Point

Every object has a magic middle to balance!

Look at these examples – a matka balanced perfectly on someone's head, a lattu spinning on its tip, and a ruler balanced on a finger.

This special spot is called the balancing point!



Gravity pulls things down, like a gentle invisible hand!

Everything around us feels this gentle pull. When you jump, gravity brings you back down. When you drop a ball, gravity makes it fall.

Gravity is always working, even though we cannot see it!

Gravity – The Invisible Puller



Gravity

Stable or Unstable? Let's See!

Which one will fall? Which one stays strong?

Look at the wide peetha stool – it sits low and wide on the ground. It's very hard to push over!

Now look at the tall tumbler – it's skinny and tall. A little push and whoops, it tips!



What do you think?

Which object would you trust to hold your favourite toy without falling?

Remember: Wide and low = Strong and stable!

More Stable or Less Stable?

Full Sandbag

A heavy sandbag sits low on the ground. Its magic middle is very close to the floor. This makes it super stable – it won't tip over easily!
Wide and low = Strong and steady!

Tall Flower Vase

A tall, thin vase stands high. Its magic middle is far from the ground. This makes it unstable – it can tip over with just a little push!
Tall and thin = Easy to tip!



low stable object



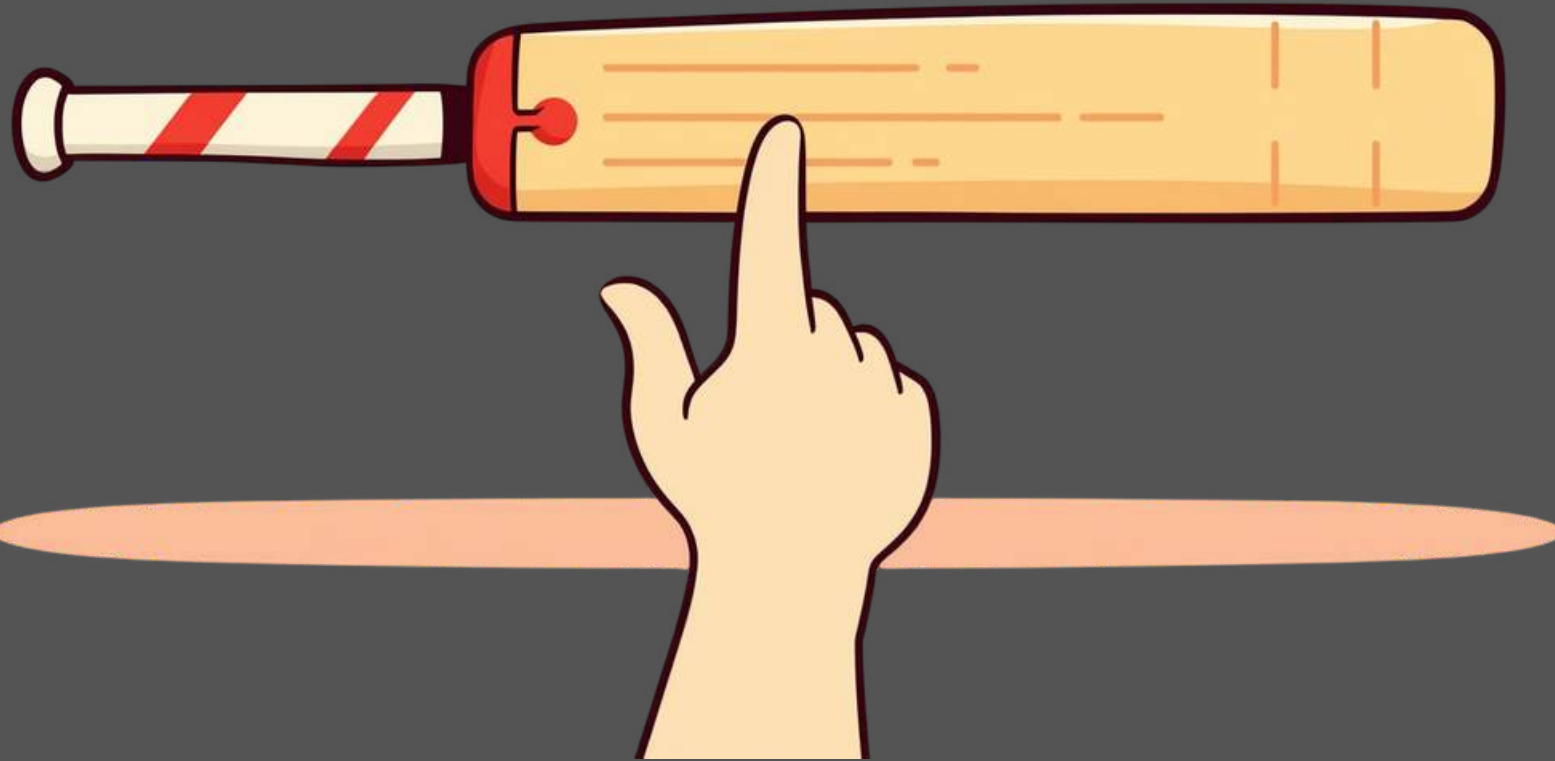
tall unstable object

Cricket Bat Balancing Act

Where is the magic middle on the cricket bat?

Try balancing a cricket bat at different points – near the handle and near the blade. Notice how it feels different!

The balancing point is closer to the blade because that's where more weight is.



Remember: Lower balancing point means more stable!

When you hold the bat by the handle, the heavy blade is up high – that's less stable. When you balance it near the blade, the weight is lower – that's more stable!



Find the Magic Middle!

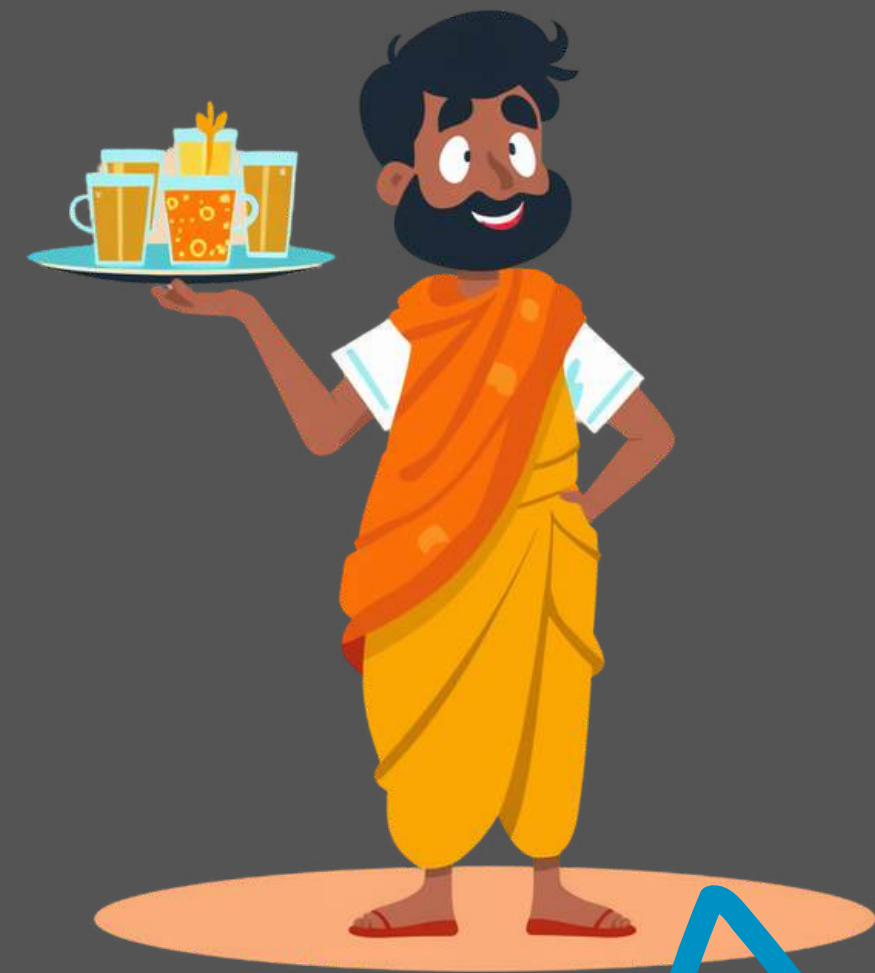


Balance a ruler on your finger. Can you find the magic middle?
Move the ruler slowly until it stays perfectly still. That's the balancing point!
Try it now! You're becoming a Balance Master!

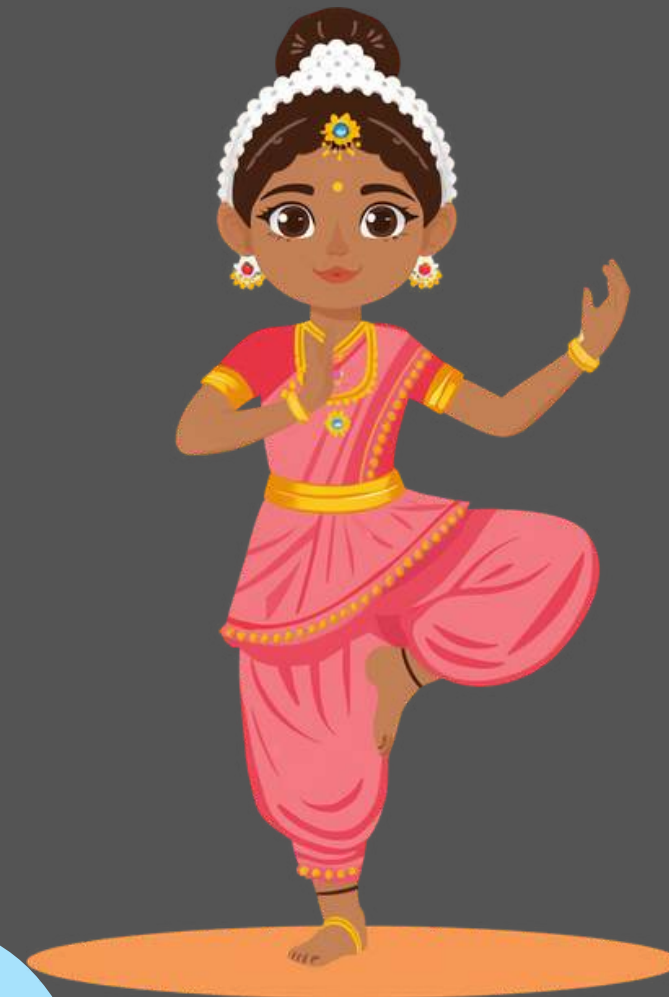
Find the Magic Middle! (Activity 2)



Can you balance a cardboard strip on a pencil tip?
Try placing the cardboard at different spots on the pencil.
Move it left, move it right!
Think about it: Is the balancing point always in the middle?



Balance All Around Us



Look around! Balance is everywhere in our daily life:

- ★ Street vendor balancing a chai tray on one hand
- ★ Bharatanatyam dancer standing on one leg
- ★ Seesaw (jhula) going up and down at the park
- ★ Stacking steel dabbas (tiffin boxes) perfectly!

Can you spot more balancing acts around you?

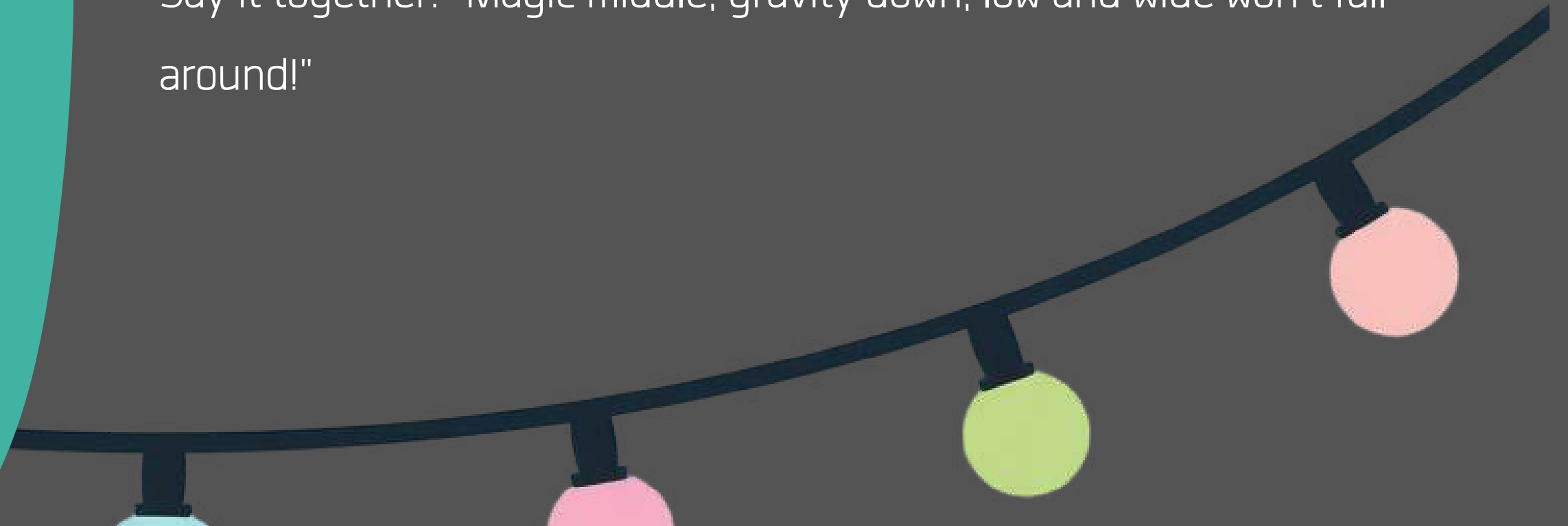


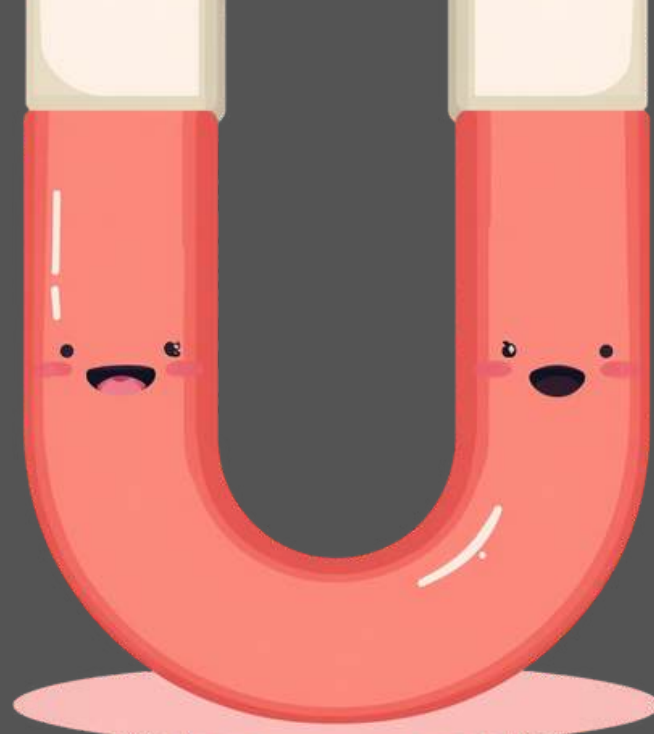
Riya Learns the Balancing Chant

- Every object has a balancing point – the magic middle where it stays steady!
- Gravity pulls everything down like a gentle invisible hand pulling towards the ground.

- Low balancing point means stable! Things with a low magic middle don't fall easily.

Say it together: "Magic middle, gravity down, low and wide won't fall around!"





Vocabulary Time!



Centre of Gravity = Magic Middle inside every object

Gravity = Invisible pull that brings things down

Stable = Doesn't fall easily, stays strong!

Unstable = Falls or tips easily, wobbly!



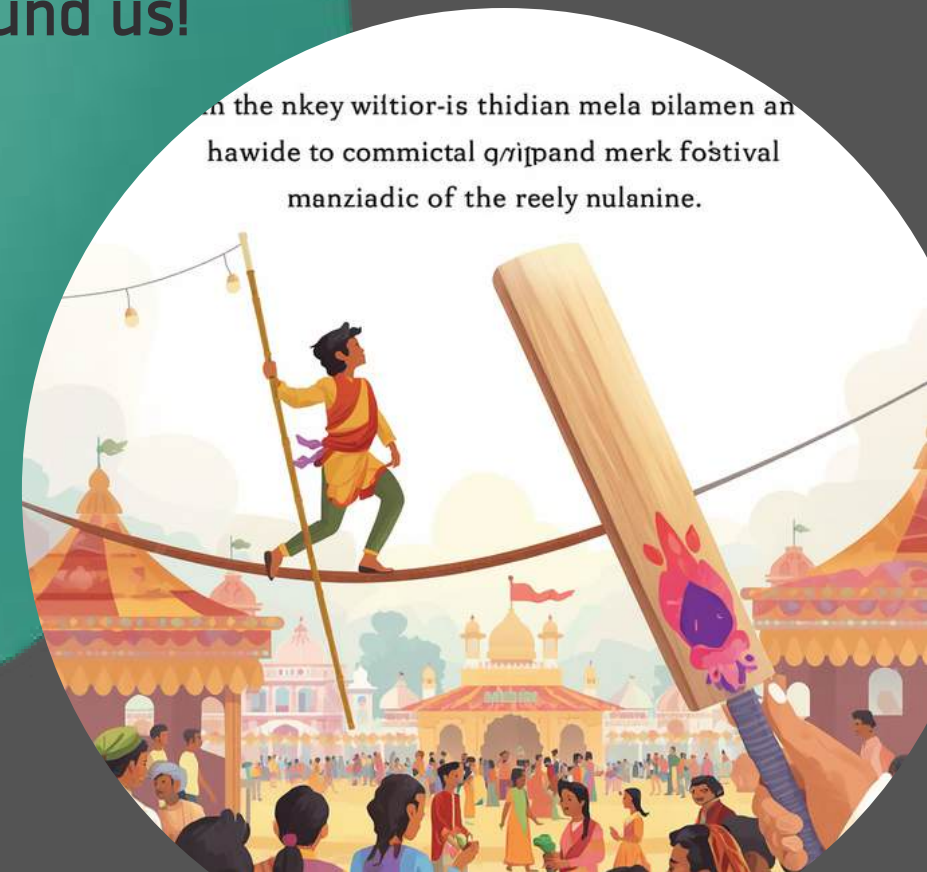
3 + 2

Indian Balance Wonders

Look at these amazing examples of balance from India! A spinning lattu stays upright on the floor. Women carry heavy matkas perfectly balanced on their heads. Tightrope walkers at the mela walk high above the ground. Cricket players hold their bats just right. Balance is everywhere around us!



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When the magic middle moves too far,
objects topple!

Imagine balancing a book on your head.
If you lean too much, the book falls! The
same happens with everything around
us.

When the balancing point goes outside
the base, things tip and fall down!

Why Things Topple Over



MATH

Stable vs Unstable Recap

Stable Objects

- ✓ Wide peetha stool – sits low and wide
- ✓ Full sandbag – heavy at the bottom
- ✓ Steel dabba stack – balanced and sturdy

These don't fall easily!



Unstable Objects

- ✗ Tall tumbler – tips over easily
- ✗ Ganesh idol – narrow base
- ✗ Empty bottle – light and wobbly

These can topple quickly!



Be the Balance Master!



Try these fun activities at home!

- ★ Balance a spoon on the rim of a glass
- ★ Try balancing a fork vs a spoon – which is easier?
- ★ Find the ruler's balancing point with a grown-up
- ★ Draw objects and mark their balancing point with a star
- ★ Find 3 stable and 3 unstable objects in your home



3+2



Wonderful! You are a
Balance Master now!
See you next time at
PLANRS ACADEMY!

